

Anil Sahith Vallepu

Portfolio: <https://anil-portfolio-chi.vercel.app>

LinkedIn: <https://www.linkedin.com/in/anil-sahith>

GitHub: <https://www.github.com/sahit1011>

Email: anilsahithvallepu@gmail.com

Mobile: +91 8143400946

PROFESSIONAL SUMMARY

Software Engineer with strong Math and CS foundations, specializing in AI/ML systems and production-grade deployment. Proven expertise in developing and optimizing deep learning models, LLM applications, and RAG systems. Skilled in building low-latency inference pipelines, on-device model optimization, and scalable data architectures.

Seeking roles: Software Engineer, AI Engineer, Machine Learning Engineer, LLM Engineer, Data Scientist.

SKILLS

- **Languages & Tools:** Python, C++, JavaScript, TypeScript, SQL, GitHub, Postman, Figma, Cursor, Codex.
- **Backend:** Next.js, Node.js, Django, Flask, FastAPI, WebSockets, ZeroMQ, RabbitMQ, Kafka, Redis Streams, TensorZero.
- **Databases & Infra:** MySQL, PostgreSQL, MongoDB, Supabase, Redis, Pinecone, ChromaDB, Modal, AWS, GCP, Docker, Kubernetes, ArgoCD, Github Actions, Prometheus, Grafana, Temporal, CubeAPM.
- **AI/ML:** TensorFlow, PyTorch, CUDA, ONNX, TensorRT, vLLM, OpenCV, LangChain, LangGraph, LLMs, VLMs, RAG.
- **Other Coursework:** Data Structures and Algorithms, System Design, Embedded Systems, AI Voice Agents.

EXPERIENCE

- **Matters.AI** Dec 2025 - Current
Software Engineer - AI Bangalore, Karnataka
 - Worked on a production-grade data processing system handling massive datasets (unstructured & structured) using Dask for distributed parallel execution and a hybrid extraction engine for PII/PHI detection, integrating VLM and LLM models for image extraction, entity classification, and post-verification.
 - Designed and deployed an inference gateway on GKE to route NER, OCR payloads across Modal and self-hosted GPU backends, with Redis Streams-based queue decoupling enabling ingestion up to 10K RPS while maintaining controlled downstream throughput up to 3k RPS. Incorporated health-aware routing, rate limiting, and resilient requeue/retry/DLQ handling, achieving ~0.001% failure rate across 1M+ documents.
 - Built high-throughput ML serving infrastructure by optimizing NER models with ONNX Runtime + TensorRT and OCR with vLLM + Flash-Attention on GPU node pools, achieving sub-100ms p99 NER batch latency and sub-500ms OCR request latency via async micro-batching and A/B tested concurrency tuning while meeting SLA and cost targets.
- **Noccarc Robotics Pvt Ltd** June 2024 - April 2025
AI Engineer Pune, Maharashtra
 - Architected a real-time arrhythmia detection system for embedded patient monitoring devices using deep learning, established an end-to-end data pipeline from sensor integration to frontend visualization with automated alarm generation across 20+ distinct classifications for critical heart abnormalities, reducing clinical response times in ICUs.
 - Implemented a hybrid neural network combining ResNet, CBAM attention mechanism, and GRU layers with separable convolutions for efficient edge deployment, achieving ~300ms latency on a patient monitor while processing ECG data every 5 seconds and consuming under 12% single-core CPU utilization.
- **Carelon Global Solutions (Elevance Healthcare)** June 2023 - August 2023
Summer Intern - ML Bangalore, Karnataka
 - Automated document processing pipeline using computer vision and deep learning to extract and classify data from 700+ documents per hour with 95% accuracy; deployed an end-to-end OCR solution using pytesseract and EasyOCR for text extraction, OpenCV for image preprocessing, and YOLOv8 object detection for region identification.
 - Accelerated processing performance through multithreading and multiprocessing techniques, reducing document processing time by 90% and decreasing operational costs, delivered a production-ready solution by integrating with existing data workflows for scalable enterprise document management.

PROJECTS

- **CryptAI - AI Trading Engine:** Built an autonomous crypto futures trading system using a multi-agent architecture where specialized AI agents analyze markets, identify swing/scalp setups, validate risk, execute real-time trades via WebSockets, store trade history, and adapt from performance, with a live dashboard for portfolio and agent activity monitoring.
Tech stack: Python, Next.js, FastAPI, Redis, PostgreSQL, LangGraph, Groq LLM API, Docker, Vercel.

EDUCATION

- **National Institute of Technology (NIT) Warangal.** Warangal, India
Bachelor of Technology in Electrical and Electronics Engineering. May 2024

ACHIEVEMENTS

- Secured 99.3rd percentile in JEE Mains (India) - Top 0.7% among 1 million+ candidates
- Recognised with Top 100 Projects at NASA AMES Space Settlement Contest, San Juan, Puerto Rico (2017)
- Solved 300+ Data Structures and Algorithms problems (Medium-Hard difficulty) on LeetCode